LDRP Institute of Technology and Research

MCA Department

Sem-1

Subject Name: Programming using open source Subject Code: MCA-15

Practical Assignment-2

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1	**Create a program that outputs the total cost of a lunch order. Users should be prompted to input the number of hamburgers, fries and drinks they want and the program should print the total cost of the order. The hamburgers cost Rs. 25/-, fries cost Rs. 50/-, and drinks cost Rs. 20/-. Be creative and professional in prompting the user for the information and displaying the output.**
Code	print("==========") print(" LUNCH ORDER BILLING SYSTEM") print("==========") hamburgers = int(input("Enter number of hamburgers (Rs.25 each): ")) fries = int(input("Enter number of fries (Rs.50 each): ")) drinks = int(input("Enter number of drinks (Rs.20 each): ")) total = (hamburgers * 25) + (fries * 50) + (drinks * 20) print("\n---------- ORDER SUMMARY ----------") print("Hamburgers :", hamburgers) print("Fries :", fries) print("Drinks :", drinks) print("----------") print("Total Cost : Rs.", total) print("==========
Output	print("==========") print(" LUNCH ORDER BILLING SYSTEM") print("==========") hamburgers = int(input("Enter number of hamburgers (Rs.25 each): ")) fries = int(input("Enter number of fries (Rs.50 each): ")) drinks = int(input("Enter number of drinks (Rs.20 each): ")) total = (hamburgers * 25) + (fries * 50) + (drinks * 20) print("\n---------- ORDER SUMMARY ----------") print("Hamburgers :", hamburgers)

	print("Fries :", fries) print("Drinks :", drinks) print("----------") print("Total Cost : Rs.", total) print("==========")
2	**Input two numbers from the keyboard and perform +, -, *, /, %, **, and // operations and print the result.**
Code	a = float(input("Enter first number: ")) b = float(input("Enter second number: ")) print("\nArithmetic Operations:") print("Addition (+) =", a + b) print("Subtraction (-) =", a - b) print("Multiplication (*) =", a * b) print("Division (/) =", a / b) print("Modulus (%) =", a % b) print("Exponentiation (**) =", a ** b) print("Floor Division (//) =", a // b)
Output	a = float(input("Enter first number: ")) b = float(input("Enter second number: ")) print("\nArithmetic Operations:") print("Addition (+) =", a + b) print("Subtraction (-) =", a - b) print("Multiplication (*) =", a * b) print("Division (/) =", a / b) print("Modulus (%) =", a % b) print("Exponentiation (**) =", a ** b) print("Floor Division (//) =", a // b)
3	**Input two numbers from the keyboard and apply comparison operators, ==, !=,** **<, >, <= and >= then print the result**
Code	a = float(input("Enter first number: ")) b = float(input("Enter second number: ")) print("\nComparison Results:") print("a == b :", a == b) print("a != b :", a != b) print("a < b :", a < b) print("a > b :", a > b) print("a <= b :", a <= b) print("a >= b :", a >= b)
Output	Enter first number: 33 Enter second number: 66 Comparision Results: a == b : False a != b : True a < b : True a > b : False

	a <= b : True a >= b : False
4	**Input three numbers from the keyboard and check their values using logical operators, 'and', 'or' and 'not'.**
Code	a = int(input("Enter first number: ")) b = int(input("Enter second number: ")) c = int(input("Enter third number: ")) print("\nLogical Operations:") print("a > 0 and b > 0 :", a > 0 and b > 0) print("a > 0 or b > 0 :", a > 0 or b > 0) print("not (c > 0) :", not (c > 0))
Output	Enter first number: 33 Enter second number: 66 Enter third number: 99 Logical Operations: a > 0 and b > 0 : True a > 0 or b > 0 : True not (c > 0) : False
5	**Write a Python program to check whether two integer variables a and b refer to the same memory location using the is operator.**
Code	a = 10 b = 10 print("a =", a) print("b =", b) print("Do a and b refer to the same memory location?") print(a is b)
Output	a = 10 b = 10 Do a and b refer to the same memory location? True
6	**Write a Python program to compare two string variables s1 and s2 using is and ==. Print the result.**
Code	s1 = input("Enter first string: ") s2 = input("Enter second string: ") print("\nComparison Results:") print("s1 == s2 :", s1 == s2) print("s1 is s2 :", s1 is s2)
Output	Enter first string: Patel Enter second string: Ritu

	Comparison Results: s1 == s2 : False s1 is s2 : False
7	**Write a Python program to check if the string "apple" exists inside the list ["banana", "apple", "grape"].**

Code	fruits = ["banana", "apple", "grape"]
	
	if "apple" in fruits:
	  print("Apple exists in the list.")
	else:
	  print("Apple does not exist in the list.")

Output	Apple exists in the list.
8	**Write a Python program to find the memory addresses of two integer variables a and b using the id() function and check whether their memory locations are different using the is not operator.**
Code	a = 10 b = 20 print("Memory address of a:", id(a)) print("Memory address of b:", id(b)) print("\nAre their memory locations different?") print(a is not b)
Output	Memory address of a: 140453764761960 Memory address of b: 140453764762280 Are their memory locations different? True
9	**Write a Python program to test whether an element 50 is not in the tuple (10, 20, 30, 40).**

Code	numbers = (10, 20, 30, 40)
	
	if 50 not in numbers:
	  print("50 is not present in the tuple.")
	else:
	  print("50 is present in the tuple.")

Output	50 is not present in the tuple.
10	**Write a Python Script to use the following mathematical functions. Take necessary inputs from the keyboard and print the result. □ abs(x) □ ceil(x)**

	□ **floor(x)** □ **exp(x)** □ **fabs(x)** □ **log10(x)** □ **min(x1,x2,x3,...,xn)** □ **max(x1,x2,x3,...,xn)** □ **modf(x)** □ **pow(x,y)** □ **round(x [,n])** □ **sqrt(x)**

Code	import math
	
	# abs(x)
	x = float(input("\nEnter a number for abs(x): "))
	print("abs(x) =", abs(x))
	
	# ceil(x)
	x = float(input("\nEnter a number for ceil(x): "))
	print("ceil(x) =", math.ceil(x))
	
	# floor(x)
	x = float(input("\nEnter a number for floor(x): "))
	print("floor(x) =", math.floor(x))
	
	# exp(x)
	x = float(input("\nEnter a number for exp(x): "))
	print("exp(x) =", math.exp(x))
	
	# fabs(x)
	x = float(input("\nEnter a number for fabs(x): "))
	print("fabs(x) =", math.fabs(x))
	
	# log10(x)
	x = float(input("\nEnter a positive number for log10(x): "))
	print("log10(x) =", math.log10(x))
	
	# min(x1, x2, x3, ..., xn)
	print("\nEnter three numbers to find minimum:")
	a = float(input("Enter first number: "))
	b = float(input("Enter second number: "))
	c = float(input("Enter third number: "))
	print("Minimum =", min(a, b, c))
	
	# max(x1, x2, x3, ..., xn)
	print("\nMaximum of the same numbers:")
	print("Maximum =", max(a, b, c))
	
	# modf(x)
	x = float(input("\nEnter a number for modf(x): "))
	fraction, integer = math.modf(x)

	print("Fractional part =", fraction)
	print("Integer part =", integer)
	
	# pow(x, y)
	x = float(input("\nEnter base x for pow(x,y): "))
	y = float(input("Enter exponent y: "))
	print("pow(x, y) =", math.pow(x, y))
	
	# round(x [,n])
	x = float(input("\nEnter a decimal number for round(x): "))
	n = int(input("Enter number of decimal places: "))
	print("round(x, n) =", round(x, n))
	
	# sqrt(x)
	x = float(input("\nEnter a positive number for sqrt(x): "))
	print("sqrt(x) =", math.sqrt(x))

Output	Enter a number for abs(x): 10 abs(x) = 10.0 Enter a number for ceil(x): 33 ceil(x) = 33 Enter a number for floor(x): 6 floor(x) = 6 Enter a number for exp(x): 67 exp(x) = 1.2523631708422137e+29 Enter a number for fabs(x): 44 fabs(x) = 44.0 Enter a positive number for log10(x): 87 log10(x) = 1.9395192526186185 Enter three numbers to find minimum: Enter first number: 33 Enter second number: 66 Enter third number: 99 Minimum = 33.0 Maximum of the same numbers: Maximum = 99.0 Enter a number for modf(x): 76 Fractional part = 0.0 Integer part = 76.0 Enter base x for pow(x,y): 4 Enter exponent y: 6

	pow(x, y) = 4096.0 Enter a decimal number for round(x): 6 Enter number of decimal places: 10 round(x, n) = 6.0 Enter a positive number for sqrt(x): 25 sqrt(x) = 5.0
11	**Write a python script to implement the following string functions.** **□ center(width, fillchar)** **□ count(str, beg, end)** **□ find(str, beg, end)** **□ isalnum()** **□ isalpha()** **□ isdigit()** **□ islower()** **□ isnumeric()** **□ isspace()** **□ isupper()** **□ len(string)** **□ lower()** **□ max(str)** **□ min(str)** **□ replace(old, new [,max])** **□ split(del_str=" ", num)** **□ swapcase()** **□ upper()**

Code	s = input("Enter a string: ")
	
	# center(width, fillchar)
	width = int(input("\nEnter width for center(): "))
	print("center() :", s.center(width, "*"))
	
	# count(str, beg, end)
	sub = input("\nEnter substring to count: ")
	print("count() :", s.count(sub))
	
	# find(str, beg, end)
	sub = input("\nEnter substring to find: ")
	print("find() :", s.find(sub))
	
	# isalnum()
	print("\nisalnum() :", s.isalnum())
	
	# isalpha()
	print("isalpha() :", s.isalpha())
	
	# isdigit()

	print("isdigit() :", s.isdigit())
	
	# islower()
	print("islower() :", s.islower())
	
	# isnumeric()
	print("isnumeric():", s.isnumeric())
	
	# isspace()
	print("isspace() :", s.isspace())
	
	# isupper()
	print("isupper() :", s.isupper())
	
	# len(string)
	print("len() :", len(s))
	
	# lower()
	print("lower() :", s.lower())
	
	# max(str)
	print("max() :", max(s) if s else "String is empty")
	
	# min(str)
	print("min() :", min(s) if s else "String is empty")
	
	# replace(old, new [, max])
	old = input("\nEnter old substring: ")
	new = input("Enter new substring: ")
	print("replace() :", s.replace(old, new))
	
	# split(del_str, num)
	print("\nsplit()")
	delimiter = input("Enter delimiter (press Enter for space): ")
	if delimiter == "":
	  delimiter = " "
	print("split() :", s.split(delimiter))
	
	# swapcase()
	print("swapcase() :", s.swapcase())
	
	# upper()
	print("upper() :", s.upper())

Output	Enter a string: Hello World Enter width for center(): 20 center() : ****Hello World*****

	Enter substring to count: l count() : 3 Enter substring to find: l find() : 2 isalnum() : False isalpha() : False isdigit() : False islower() : False isnumeric(): False isspace() : False isupper() : False len() : 11 lower() : hello world max() : r min() : Enter old substring: World Enter new substring: Python replace() : Hello Python split() Enter delimiter (press Enter for space): split() : ['Hello', 'World'] swapcase() : hELLO wORLD upper() : HELLO WORLD